Brian Chu

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WORK EXPERIENCE

Scale AI | San Francisco, CA **April 2026 - Present**

Forward Deployed AI Engineer

- Shipped 20+ structured-output features to enterprise health clinic and engineering firm clients by building internal MCP tooling with a deterministic architecture that plugs into AI agents and evals
- Recovered $10M in credits across 15K+ documents by deploying an AI extraction pipeline orchestrated in Temporal, routing low-confidence extractions to human-in-the-loop (HITL) reviewers
- Informed a $2M budget reallocation by building executive metrics dashboards on 1M+ weekly logged events, replacing manual reports with real-time visibility into credit recovery

Software Engineer

- Cut new feature build time from 2 weeks to 3 days by reusing shared, deterministic MCP tooling with Redis caching across client projects instead of building each integration from scratch
- Caught agent regressions before client releases by scheduling 1,000+ daily eval cases against AI agent outputs with GCP Cloud Run jobs and Cloud Scheduler for enterprise deployments
- Automated trade research with a multi-agent portfolio trading system of tool-calling research and strategy agents, gated by deterministic risk checks, AWS IAM-scoped access, and HITL approval

Stealth Startup | Remote **May 2022 - April 2026**

AI Engineer

- Replaced manual machinist quoting with autonomous pricing across 7 cost categories by deploying tool-calling agent loops over 3 LLM providers with Pydantic output via FastAPI microservices
- Achieved 40% higher engineering drawing analysis accuracy by developing a VLM-based preprocessing pipeline using Gemini Vision for classification, assembly trees, and BOM extraction
- Cut pricing lookup latency 70% by consolidating three quote workflows into unified React, FastAPI, and PostgreSQL microservices with recursive BOM quoting and cached OAuth vendor pricing APIs

Jacobs School of Engineering, UCSD | La Jolla, CA **January 2023 - June 2023**

Thermal Process Engineer

- Achieved 30% higher energy efficiency and 95% methanol conversion by designing an industrial-scale dimethylamine reactor, heat exchanger system, and P&ID in Aspen Plus & COMSOL

PROJECTS

Speed Demon - F1 AI Racing Technology Automation **September 2025 - Present**

- Engineered real-time telemetry system using OpenTelemetry, with a time-series database on AWS EC2, integrating LangChain agents utilizing skills and MCP with GraphRAG to predict driver style
- Orchestrated a distributed data processing backend by implementing batch and stream processing using PySpark on AWS EC2 cloud infrastructure with centralized monitoring

NLP Drug-Drug Interaction Prediction System **May 2025 - August 2025**

- Predicted combined drug effects with 82% accuracy by developing a Knowledge Graph Neural Network for 10K+ drug-drug interaction descriptions with TensorFlow and Neo4j
- Processed pharmacological text data by tokenizing with spaCy and building feature pipelines with Hugging Face Transformers, improving model input quality for the Knowledge Graph Neural Network

EDUCATION

University of California, San Diego | San Diego, CA

Master of Science in Chemical Engineering, FE Chemical **August 2024 - December 2025**

Bachelor of Science in Computer Science and Chemical Engineering **August 2020 - June 2024**

SKILLS

Languages | Frameworks: Python | TypeScript | C++ | SQL | LangChain | TensorFlow | FastAPI | Pydantic

Tools | Infrastructure: AWS | GCP | Kubernetes | Docker | Temporal | PostgreSQL | Redis | CI/CD